

Google Cloud Global Oil & Gas Industry Strategy

Executive Strategy & Market Execution Roadmap Google Cloud · Global Oil & Gas Industry Team August 2026 · STRATEGIC EXECUTIVE BRIEFING · CONFIDENTIAL

Executive Summary

The global oil and gas industry is entering a period of unprecedented convergence. Five structural forces — the AI compute inflection, a multi-decade data fragmentation crisis, an aging technical workforce, mandatory sovereign data residency laws, and the accelerating energy transition — are simultaneously reshaping how the world's largest energy operators invest in technology. For Google Cloud, this convergence represents a generational opportunity to redefine our position in one of the world's most capital-intensive verticals.

Today, the oil and gas sector generates **\$4.9 trillion in annual revenue** and spends **\$44 billion annually on IT and digital infrastructure**, yet only **~18% of O&G workloads run on public cloud** (IEA World Energy Outlook 2025; Gartner IT Spending Forecast Q2 2025; IDC Cloud Tracker 2025). Google Cloud currently holds approximately **10% market share** in this vertical, trailing AWS at 37% and Microsoft Azure at 32%. This document presents a strategy to **double our market share to 20% within three years**, capturing a **\$3.2 billion serviceable obtainable market** from a total addressable market of **\$21 billion** by 2028.

This is not a strategy built on competing symmetrically against AWS and Azure. It is built on four asymmetric advantages that only Google Cloud possesses:

Gemini's long-context multimodal reasoning — the ability to ingest entire 30-year well log repositories, seismic surveys, and operational histories into a single 2-million-token prompt without chunking loss — a capability no competitor can match today.

The Alphabet portfolio — DeepMind's frontier science engine, Google Earth Engine's unmatched satellite catalog, Tapestry's grid intelligence, and Google's own industrial-scale energy procurement organization — none of which AWS or Azure can replicate.

The multicloud co-existence wedge — BigQuery Omni and BigLake's ability to query data in-place across AWS S3 and Azure ADLS without requiring data migration — removing the single largest objection in every competitive displacement.

The Energy Exchange — Google's position as simultaneously the world's largest AI infrastructure builder *and* one of the world's largest energy buyers — creating bilateral deal structures that no other technology company can offer.

This strategy is organized around **six operating pillars**, each designed to compound the others, and executed through a disciplined **90-day phased roadmap** with measurable exit criteria at each gate.

Part I: The Industry Landscape — Why This Market, Why Now

1.1 The Scale of the Opportunity

The oil and gas industry is not merely large — it is one of the most technology-intensive sectors in the global economy, and it is dramatically underserved by cloud providers. To understand the opportunity, consider the following market dimensions:

The industry's **\$4.9 trillion annual revenue** is generated by operators managing some of the most complex physical systems on Earth — deepwater drilling platforms operating at pressures exceeding 20,000 PSI, LNG liquefaction trains processing millions of tonnes annually, pipeline networks spanning tens of thousands of kilometers, and refineries running continuous chemical processes that have not been shut down in decades. Each of these operations generates massive volumes of sensor data, geological data, operational data, and regulatory data — yet the vast majority of this data remains trapped in disconnected silos ([IEA World Energy Outlook 2025](#)).

The sector's **\$44 billion in annual IT spending** is growing at approximately 6-8% annually as operators invest in digital transformation, but **cloud penetration remains at only ~18%**, significantly below the cross-industry average of 40-45% ([Gartner IT Spending Forecast Q2 2025](#)). This gap represents the single largest cloud adoption opportunity in any major industrial vertical.

Within this broader market, two sub-segments are growing at exceptional rates. The **AI and machine learning market in oil and gas** is valued at **\$5.4 billion today** and is projected to reach **\$18.7 billion by 2035**, growing at a 13% CAGR ([Precedence Research 2025](#)). The **digital oilfield solutions market** stands at **\$37 billion** and is projected to reach **\$43 billion by 2029**, driven by the adoption of IoT, edge computing, and predictive analytics across field operations ([MarketsandMarkets 2025](#)).

Meanwhile, the **CCUS (Carbon Capture, Utilization, and Storage) investment pipeline** has surpassed **\$5 billion annually**, a 15× increase since 2020, with 77 CCUS facilities now operating globally and 47 additional projects under construction ([Global CCS Institute Status Report 2025](#)). This represents an entirely new technology platform market that did not exist at meaningful scale five years ago — and one where no hyperscaler has yet established dominance.

1.2 Five Structural Forces Driving Urgency

The reason this opportunity is actionable *now* — rather than being a long-term horizon play — is the simultaneous convergence of five structural forces that are compelling operators to make technology platform decisions in the next 12–24 months.

Force 1: The AI Inflection Point. Energy companies find themselves on both sides of the AI revolution. As operators, they need AI to optimize production, reduce non-productive time, predict equipment failures, and accelerate subsurface interpretation. As energy suppliers, they are the providers of the firm, dispatchable power that AI data centers require. AI-driven workflows are already delivering **20% efficiency gains** in early-adopter operations, and data centers are projected to consume as much electricity as entire nations by 2030 ([IEA Electricity 2024](#)). This dual positioning creates a unique bilateral commercial opportunity that we exploit through Project Interchange (detailed in Part IV). **Force 2: The Data Fragmentation Crisis.** The average major operator manages **20–30 years of siloed data** spread across OSDU data platforms, PI/OSIsoft historians, legacy paper well logs, SAP and Oracle ERPs, Petrel and Techlog interpretation environments, and dozens of custom databases. The economic cost of this fragmentation — in duplicated analysis, missed correlations, slow decision cycles, and regulatory non-compliance — has escalated from an IT inconvenience to a **board-level risk** ([SPE Digital Energy Conference 2025](#)).

Operators are now actively seeking platforms that can reason across these silos without requiring the politically impossible task of consolidating them — precisely what our Gemini-powered agentic architecture delivers. **Force 3: The Workforce Aging Crisis.** The average age of an upstream geoscientist now exceeds **50 years**, and a significant share of the experienced technical workforce is expected to retire by 2030 ([SPE/AAPG Workforce Surveys 2024–2025](#)). These are the professionals who carry decades of interpretive judgment about specific basins, formations, and operating environments — knowledge that cannot be replaced by hiring alone. Agentic AI that can encode, preserve, and augment this institutional knowledge is not an efficiency play; it is a workforce continuity strategy that boards are beginning to treat as existential. **Force 4: Sovereign AI Mandates.** National oil companies in Saudi Arabia, Qatar, Kuwait, Indonesia, Thailand, and Japan are subject to increasingly strict data residency laws that mandate in-country processing of sensitive operational data. Saudi Arabia's NCA regulations, Indonesia's GR-71, Thailand's PDPA, and Japan's sovereign AI guidelines all create hard requirements for in-country cloud infrastructure ([Saudi NCA Regulations](#)). These requirements differentiate cloud providers sharply — and Google Cloud's sovereign region portfolio (Dammam, Jakarta, Bangkok, Tokyo, and GDC Air-Gapped) positions us to serve NOC workloads that neither AWS nor Azure can fully address today. **Force 5: Energy Transition and CCUS.** Clean energy investment surpassed fossil fuel investment for the first time in 2025. Operators must simultaneously grow production to meet near-term demand *and* decarbonize their operations to meet net-zero commitments. CCUS project pipelines have grown at **30%+ CAGR since 2017**, creating a new category of technology-intensive operations — CO₂ capture, transport, injection monitoring, and regulatory compliance — that require purpose-built digital platforms ([Global CCS Institute Status Report 2025](#)). The CCUS platform market is currently unowned by any hyperscaler, and our SLB partnership positions us to claim it.

1.3 Market Sizing: TAM, SAM, and SOM

We structure our addressable market across three concentric tiers:

The **Total Addressable Market (TAM)** of **\$21.0 billion by 2028** encompasses all cloud, data, and AI infrastructure spending across upstream, midstream, and downstream oil and gas operations globally. This figure is derived from IDC Cloud Tracker and Straits Research projections for energy vertical cloud adoption ([IDC Cloud Tracker 2025](#)).

The **Serviceable Addressable Market (SAM)** of **\$9.5 billion** represents the subset of workloads where Google Cloud has a differentiated right to win: agentic AI and reasoning workloads (where Gemini's context window is decisive), sovereign NOC deployments (where our regional footprint is strongest), subsurface HPC computing (where TPU economics are superior), and CCUS digital platforms (where our SLB partnership is unique). This SAM excludes commodity IaaS workloads where we have no structural advantage ([Gartner O&G Vertical Cloud Analysis 2025](#)).

The **Serviceable Obtainable Market (SOM)** of **\$3.2 billion** represents the realistic three-year capture from our 33 named accounts, calibrated against pipeline maturity, competitive entrenchment, and execution capacity.

1.4 Cloud Market Share and Competitive Position

Understanding where we stand competitively is essential to understanding why a different strategy is required. Today's market share distribution in the O&G vertical reflects historical enterprise relationships and migration-era decisions, not current technical superiority:

AWS holds 37% market share, anchored by deep relationships with Shell, Oxy, Aker BP, TGS, ENI, and Petrobras. AWS's strength is infrastructure scale and breadth of services, but its AI capabilities lag Google's in reasoning depth, multimodal context, and custom silicon economics. **Microsoft Azure holds 32%**, anchored by Chevron, Equinor, Devon, ADNOC, and Petronas. Azure's strength is enterprise integration (Office 365, Teams, Dynamics)

and existing Microsoft EA agreements, but its multicloud federation capabilities are weaker than Google's, and its sovereign cloud offerings in the Gulf and APAC are less mature. **Google Cloud holds 10%**, with anchor relationships including Pertamina, PTTEP, Reliance Industries, and TotalEnergies (via SLB). Our position is smaller but strategically positioned — we hold the sovereign APAC franchise, the SLB technology partnership, and the most advanced AI reasoning platform in market. **Our three-year target is 20% market share**, which translates to approximately **\$1.96 billion in annual recurring revenue** by 2028 across our named account portfolio.

1.5 Revenue Growth Trajectory

Our revenue projection model, built bottom-up from named account pipeline analysis and validated against Gartner and IDC growth forecasts, projects the following five-year trajectory across five workload categories (figures in \$ millions):

Year	AI/ML	HPC Seismic	Data & OSDU	Sovereign	CCUS	Total
2024	80	40	120	30	5	275
2025	140	65	180	55	15	455
2026	250	110	260	95	40	755
2027	450	180	380	160	80	1,250
2028	750	280	520	260	150	1,960

Source: Directional model based on Gartner IT Spending Forecast, IDC Cloud Tracker, and named account pipeline analysis.

The steepest growth curves are in **AI/ML workloads** (75% CAGR, driven by Gemini adoption and agentic OSDU deployments) and **CCUS** (134% CAGR, driven by regulatory mandates and our SLB partnership). **Sovereign cloud** (71% CAGR) reflects the expanding NOC pipeline as Dammam, Doha, and APAC regions reach feature parity.

1.6 Digital Maturity Landscape

Not all operators are equally ready to adopt cloud and AI at scale. Our analysis of the 33 named accounts reveals a four-stage maturity distribution that informs our engagement sequencing:

Leaders (Digital Maturity Score 85-92): TotalEnergies, Shell, BP, and Equinor have advanced digital organizations, significant cloud footprints, and active AI programs. These operators are sophisticated buyers who evaluate on technical differentiation, not marketing. Our strategy with leaders is the multicloud wedge — proving value on top of their existing AWS/Azure estates without requiring migration. **Fast Followers** (Score 65-78): Saudi Aramco, ExxonMobil, Pertamina, Devon Energy, EQT/Expand, and Reliance have committed to digital transformation but are still making platform decisions. These operators are receptive to new technology partnerships, particularly when framed around sovereign requirements (Aramco, Pertamina) or operational imperatives (Devon's merger integration, EQT's production optimization). **Early Adopters** (Score 45-55): KOC/KPC, QatarEnergy, Diamondback, ONGC, and PEMEX are in earlier stages of cloud adoption with significant greenfield opportunity. Our strategy here is to establish Google Cloud as the primary platform before competitors gain incumbency. **Nascent** (Score 30-35): Continental Resources, YPF, and Mewbourne Oil have minimal cloud presence. These accounts represent future pipeline but are not Day 1 priorities. *Maturity scores are directional estimates based on public digital transformation disclosures, analyst reports (Gartner/IDC), and SPE conference publications.*

Part II: The Six-Pillar Operating Strategy

Our strategy is organized around six interlocking pillars. Each pillar is designed to be independently valuable but to compound the effects of the others — a customer won through sovereign cloud (Pillar 3) becomes a candidate for the Energy Exchange (Pillar 5); an ISV partnership (Pillar 2) generates pipeline for the account offense (Pillar 1); a DeepMind research engagement (Pillar 6) creates executive relationships that accelerate sovereign deals.

Pillar 1: Customer Offense — The 33-Account Attack

The foundation of this strategy is a disciplined, scored account model. Rather than pursuing the oil and gas vertical broadly, we concentrate resources on **33 named accounts** segmented into five tiers based on strategic value, competitive position, and execution readiness.

Tier 1A — High-Velocity Independents (EQT/Expand Energy, Devon Energy, Diamondback, Harbour Energy, TGS, EOG Resources, ConocoPhillips, Aker BP): These are mid-to-large independents where Google Cloud can establish itself as the primary platform. EQT is the largest US natural gas producer and a prime candidate for our Energy Exchange framework; Devon is undergoing a major merger integration that creates a natural cloud platform decision point; Diamondback offers a clean greenfield in the Permian Basin; TGS is an active AWS win-back target experiencing GPU capacity failures on AWS. Each Tier 1A account has a named P1 executive sponsor and a defined attack playbook. **Tier 1B — PE-Backed Sponsors** (Quantum Capital Group, EnCap Investments): Private equity sponsors represent a platform-level opportunity — winning the sponsor relationship unlocks portfolio-wide replication across 15-20 portfolio companies. Quantum Capital alone manages assets across multiple Permian Basin operators. **Tier 1C — Large Privates** (Continental Resources, Mewbourne Oil): Large private operators with minimal cloud presence offer greenfield platform opportunities unconstrained by existing enterprise agreements. **Tier 2 — Sovereign NOCs** (Saudi Aramco, KOC/KPC, Pertamina, PTTEP, Inpex, QatarEnergy, Petronas, ADNOC, ONGC): National oil companies represent the highest-value individual account opportunities in the portfolio. Saudi Aramco alone represents an estimated **\$150 million opportunity** over three years. These accounts are won on sovereign compliance, in-country delivery, and national AI partnership — not on feature comparison or pricing alone. Our sovereign region portfolio (Dammam, Jakarta, Bangkok, Tokyo) gives us a structural advantage that AWS and Azure cannot easily replicate. **Tier 3 — Affinity Majors** (TotalEnergies, Reliance Industries, ExxonMobil, Repsol, ENI, BP, Petrobras, Woodside, Hess, PEMEX, YPF, Santos): Global-scale operators where Google Cloud pursues either a lead position (TotalEnergies via SLB, Reliance via existing GCP anchor) or a wedge position (ExxonMobil, BP, ENI via multicloud in-place analytics). **Tier 4 — Midstream Infrastructure** (Williams Companies): Midstream operators control pipeline networks, processing plants, and gathering systems that generate massive IoT data streams. Williams already runs Kongsberg Kognitwin digital twin on GCP, creating an expansion pathway. **Fortress Accounts — Co-Existence Wedge Doctrine** (Shell, Oxy, Chevron, Equinor): These accounts are deeply entrenched with AWS or Azure through multi-year enterprise agreements. Our strategy here is explicitly *not* displacement — it is co-existence. We position BigQuery Omni and BigLake for in-place federated queries over their existing S3/ADLS data estates, Gemini for reasoning over their existing data without migration, and targeted HPC burst for seismic workloads where our TPU economics are superior. The goal is to establish a growing GCP footprint alongside the incumbent, earning expansion through demonstrated value rather than forced migration.

Pillar 2: Ecosystem Coalition — Partners That Outsell the Direct Force

No cloud provider can serve the oil and gas vertical alone. The technical complexity of upstream operations, the domain specificity of subsurface interpretation, and the regulatory diversity across 30+ operating jurisdictions require a deep partner ecosystem. Our ecosystem strategy is organized around three layers:

Layer 1 — Domain ISVs: Cognite (Data Fusion industrial data platform), SLB (Delfi/Sequestri subsurface and CCUS), Baker Hughes (Cordant IET), Kongsberg Digital (Kognitwin digital twin), and AspenTech (refinery optimization). These ISVs bring domain credibility and installed base access that Google cannot replicate independently. Our engagement model with each ISV is a signed Joint Business Plan (JBP) with co-sell targets, joint solution architectures, and shared pipeline. **Layer 2 — Global System Integrators:** EPAM (OSDU and agentic delivery), Accenture (scale enterprise delivery), and TCS/Infosys/Wipro (APAC sovereign delivery). GSIs are the multiplier force — they enable us to serve 33 named accounts simultaneously without scaling our direct team proportionally. Our target is **≥30% of qualified pipeline partner-originated or co-sold** by Day 90. **Layer 3 — Sovereign & Innovation Partners:** HUMAIN AI (Saudi Arabia sovereign AI enablement on Dammam me-central2), CNTXT (Saudi sovereign cloud operations), and our Google for Startups Cloud Program cohort (Highwood Emissions, Qube Technologies, Carbon Direct, Planet/MethaneSAT, Corva, ResFrac, Subsurface AI) receiving up to \$200,000 in GCP credits, Vertex AI grants, and accelerated Marketplace onboarding. Additionally, we are pursuing **industry foundation model co-development** with six ISV data partners — TGS, SLB, Baker Hughes, Siemens, AspenTech, and Enverus — to build GCP-hosted domain foundation models (Timeseries FM for production/drilling operations and Subsurface FM for seismic/reservoir intelligence). The IP framework is clean: zero joint IP between Google and ISVs, with joint IP reserved exclusively for tripartite engagements that include customer/operator participation, and operator-customized fine-tuned weights belonging exclusively to the operator.

Pillar 3: Technology Supremacy — The Agentic Stack and Sovereign Architecture

Our technology strategy is built on three interconnected capabilities that create a defensible moat:

Capability 1: Agentic OSDU and Beyond-OSDU Intelligence. The oil and gas industry has spent a decade building OSDU (the Open Subsurface Data Universe) as a shared data standard. However, the reality is that most operator data remains *outside* OSDU — in PI historians, legacy databases, paper archives, and vendor-specific formats. Our agentic architecture uses Gemini's 2-million-token multimodal context window to reason across both OSDU-compliant and non-OSDU data sources simultaneously, without requiring data consolidation. We define six reference agent patterns: Subsurface Interpretation Agent, Well Log Correlation Agent, Drilling Risk Agent, Production Optimization Agent, Reliability/Maintenance Agent, and Process Safety Agent. Each pattern has defined input sources, reasoning chains, governance controls (human approval gates, kill switches, audit logging), and output actions. **Capability 2: Subsurface HPC Supercomputing.** Seismic processing — specifically Reverse Time Migration (RTM) and Full Waveform Inversion (FWI) — remains the most compute-intensive workload in the energy sector. Our A3 Mega/Ultra GPU clusters (NVIDIA H100/H200/B200) with 3.2 Tbps GPUDirect RDMA networking, combined with Parallelstore (DAOS-based) sub-millisecond storage I/O and hybrid cloud bursting via the Google Cloud HPC Toolkit, deliver a **meaningfully lower total cost of ownership** compared to AWS EC2 UltraClusters with FSx for Lustre. This is not a marginal improvement — it is the difference between operators running their seismic reprocessing jobs on a cloud provider's terms versus their own. **Capability 3: The Sovereign Trio.** Our sovereign cloud architecture is organized as a unified decision framework across six deployment options, each mapped to specific regulatory requirements and operator needs:

- **Dammam (me-central2):** Class C CST/NCA certified; CNTXT-operated External Key Management with Key Access Justifications; KSA Data Boundary. Target accounts: Saudi Aramco, CNTXT, HUMAIN AI.
- **Doha Region:** In-country Qatar data boundary with low-latency regional AI inference. Target accounts: QatarEnergy, Gulf LNG ecosystem.
- **Jakarta (asia-southeast2):** In-country Indonesia data residency aligned with Pertamina Digital Hub IT Strategic Plan 2025–2029.

- **Bangkok (asia-southeast1):** In-country Thailand data residency; PTTEP already running Apigee/BigQuery/GKE on GCP.
- **Tokyo / Osaka:** In-country Japan data residency with dual-region resilience for Inpex Ichthys LNG and CCUS monitoring.
- **GDC Air-Gapped:** Google Distributed Cloud hardware deployed on-premise at critical OT sites — fully disconnected from the public internet. Target: Kuwait NOCs, ADNOC OT/ICS environments.

The critical strategic point is that we present this as **one sovereign framework with six deployment models** — not six separate products. NOC CIOs need a single conversation about sovereignty, not six separate procurement cycles.

Pillar 4: Alphabet Advantage — Capabilities No Competitor Can Replicate

Google Cloud's most underutilized strategic asset is the broader Alphabet portfolio. AWS is a standalone cloud business. Azure is a cloud business integrated with Microsoft's productivity suite. Google Cloud is part of an organization that includes DeepMind (the world's leading AI research lab), Google Earth Engine (40+ years of satellite imagery with native ML), Tapestry (an Alphabet X moonshot for grid intelligence), Mandiant/Wiz (world-class cybersecurity), and one of the world's largest clean energy procurement organizations. No competitor can assemble this combination.

Google DeepMind provides operational weather risk prediction (WeatherNext), materials discovery (GNoME — 2.2 million new materials discovered), molecular simulation (AlphaFold 3), and differentiable physics simulation (TORAX). These are not abstract research capabilities — they solve concrete energy industry problems: CCUS solvent design, pipeline corrosion alloy discovery, reservoir proxy simulation, and autonomous drilling control.

Google Earth Engine Enterprise provides 40+ years of satellite imagery paired with native ML for automated methane plume detection, pipeline right-of-way monitoring, flaring audits, and CCUS storage site selection. It powers the UNEP International Methane Emissions Observatory and EDF MethaneSAT — no competing satellite analytics platform has comparable depth. **Mandiant and Chronicle** provide OT/ICS threat intelligence and zero-trust monitoring purpose-built for operational technology environments — drilling rigs, refinery control systems, and pipeline SCADA networks. **Tapestry** (Alphabet X) provides AI-powered grid planning and interconnection intelligence — directly relevant to operators involved in power generation, grid-adjacent energy production, and the Energy Exchange. **Google Energy Procurement** is the organization responsible for Google's massive clean energy portfolio — including the \$4.75 billion acquisition of Intersect Power and a 200MW fusion PPA with Commonwealth Fusion Systems. This is not a vendor relationship; it is a peer-to-peer energy business conversation that no other cloud provider can convene.

Pillar 5: Energy Exchange — The Bilateral Operating System

Project Interchange is the most strategically differentiated initiative in this plan — and the one most likely to create category-defining outcomes. The thesis is straightforward:

Google is spending **\$195–205 billion in CapEx in 2026** with **\$811 billion in contracted future infrastructure spending** — the largest infrastructure build in history. The dominant cost driver is power. Energy operators hold the world's largest portfolio of firm, dispatchable, 24/7 power assets. **No other hyperscaler is simultaneously the world's largest AI company AND a company that needs to buy energy at industrial scale.**

The mechanism is a formal bilateral framework: operators provide firm power generation (natural gas, geothermal, nuclear SMR, battery storage) for Google data centers, and in return gain preferential access to the full Alphabet AI stack at anchor pricing tiers — including DeepMind science, Tapestry grid planning, WeatherNext, and Gemini

autonomous operations. Google Energy Procurement leadership joins operator executive sessions as a second badge at the table — creating a meeting that no competitor can convene.

Our initial pilot accounts for Interchange are:

- **EQT / Expand Energy:** The largest US natural gas producer with Fervo geothermal adjacency. Google needs firm gas-fired and geothermal power for US data centers; EQT needs AI-native production optimization across its Appalachian Basin assets.
- **Pertamina:** Operator of the world's largest geothermal portfolio (2GW+) and an existing GCP customer. Google needs APAC clean power for the Jakarta region; Pertamina needs sovereign AI across its six sub-holdings.
- **TotalEnergies:** 40GW global renewable and gas portfolio and Europe's largest energy trader. Google needs European and global data center power; TotalEnergies needs cross-asset AI via Cognite on GCP.

Pillar 6: THINK BIG — Three Initiatives Only Alphabet Can Execute

Beyond the core commercial strategy, we propose three transformative initiatives that exploit the full Alphabet portfolio in ways that no other technology company can replicate:

Initiative 1: Project Interchange (described above as Pillar 5) — the Energy-AI Bilateral Operating System.

Initiative 2: DeepMind Energy Lab. The energy industry spends over **\$80 billion annually on R&D**, yet fundamental bottlenecks — CCUS solvent design, corrosion-resistant alloys, catalytic optimization, reservoir simulation — remain locked behind decades-old experimental approaches. Google DeepMind is the only AI lab that has simultaneously solved protein folding (AlphaFold), discovered 2.2 million new materials (GNoME), built an automated science lab for physical synthesis, created a differentiable plasma simulator (TORAX), and committed \$40 million to the DOE Genesis Mission. We propose a structured joint research program — the DeepMind Energy Lab — where Google partners with energy majors, NOCs, and national labs to apply these frontier capabilities to the hardest unsolved problems in energy. This converts Google from a cloud vendor into the energy industry's indispensable R&D partner — creating generational lock-in, not transactional cloud consumption.

Research verticals include: **Molecular discovery for energy transition** (GNoME + AlphaFold screening 100,000+ candidate molecules for CO₂ binding affinity and corrosion resistance), **physics simulation at frontier scale** (differentiable neural proxy simulators running 10,000× faster than legacy reservoir solvers), and **autonomous operations science** (reinforcement learning control policies for drilling, production allocation, and process safety).

Initiative 3: CCUS Transformation Partnerships. The world's largest CCUS mega-consortiums — ExxonMobil's 9 MTPA Gulf Coast Hub, the UK East Coast Cluster (BP/Equinor/TotalEnergies with 1 billion tonnes of storage), and Project Greensand (INEOS-led, the EU's first offshore CO₂ storage) — currently have **no dedicated cloud or AI technology partner**. Microsoft has locked Northern Lights; AWS has locked Occidental. But the majority of the CCUS market is unplatformed. Our differentiated stack — Earth Engine (80+ petabytes of satellite data), MethaneSAT (3–8 meter resolution emissions attribution), Gemini Agentic AI (A2A autonomous operations), OSDU v1.0, and SLB Delfi/Sequestri running natively on GCP — makes Google uniquely positioned to become the end-to-end AI platform across the full CCUS lifecycle. Each consortium win becomes a wedge into the operator's broader enterprise cloud portfolio — CCUS is the pathway to enterprise-wide relationships with ExxonMobil, BP, Equinor, TotalEnergies, and INEOS.

Part III: Competitive Intelligence

3.1 Head-to-Head Capability Analysis

Our competitive advantage is not uniform — it is concentrated in specific capability domains where our technical architecture creates structural differentiation. The following analysis identifies where we lead, where we are at parity, and where we must compete asymmetrically:

Where We Lead Decisively: *Generative AI Reasoning Depth.* Gemini 3.5 Pro's native 2-million-token multimodal context window allows entire 30-year well log repositories, high-resolution seismic images, and operational document histories to be ingested into a single prompt. AWS Bedrock supports up to 1M tokens via Claude but most native models are limited to 128k–200k tokens. Azure OpenAI supports up to 1M tokens via GPT-4.1 but standard models are 128k. This 2× context advantage is decisive for subsurface interpretation workloads where chunking and retrieval collapse under complex geological reasoning. *Multicloud In-Place Analytics.* BigQuery Omni and BigLake enable federated query execution over data residing in AWS S3 and Azure ADLS — without data movement, egress charges, or migration risk. No equivalent capability exists on AWS (Redshift Spectrum requires data staging) or Azure (Synapse Link is constrained). This capability is the foundation of our entire fortress account strategy. *Custom AI Silicon Economics.* Google's 6th-generation Trillium (v6e) and 5th-generation TPU v5p chips offer up to 50% TCO savings for optimized inference workloads compared to GPU-only providers (workload-dependent; CUDA migration costs may apply). AWS Inferentia/Trainium has limited O&G support. Azure offers GPU-only compute at high cost with strict quotas. *Geospatial Satellite AI.* Google Earth Engine's 40+ years of satellite imagery with native ML has no equivalent. AWS Ground Station requires custom pipeline assembly with SageMaker. Azure Orbital Analytics is growing but has a limited energy ML catalog. **Where We Must Compete Asymmetrically:** *Enterprise IT Integration.* Azure holds a structural advantage through Microsoft 365, Teams, and existing Enterprise Agreements. We do not attempt to compete on enterprise productivity tooling — instead, we compete on AI reasoning depth and multicloud federation, which are orthogonal to the Microsoft EA relationship. *Installed Base Scale.* AWS's 37% and Azure's 32% share represent years of accumulated migrations and enterprise relationships. We do not compete on base IaaS — we compete on incremental AI, HPC, and sovereign workloads where our technology is differentiated and the incumbent relationship is irrelevant to the buying decision.

3.2 Account-Level Competitive Positioning

For each of the 33 named accounts, we have developed a specific competitive positioning that reflects the incumbent's strength and our entry strategy. The general framework:

Lead accounts (Greenfield or GCP Anchor): Position Google Cloud as the primary platform. No incumbent displacement required. Strategy: speed to production. **Wedge accounts** (AWS or Azure incumbent): Position BigQuery Omni/BigLake for in-place analytics over the incumbent's data estate. No migration messaging. Strategy: prove value on a bounded use case, then expand. **Co-Exist accounts** (Deep AWS/Azure fortress): Accept co-existence as the explicit, sanctioned strategy. Position for targeted workloads (HPC burst, AI reasoning, geospatial) that complement rather than replace the incumbent. Strategy: earn the right to expand over 18–36 months.

Part IV: Execution Roadmap — 90 Days in Three Phases

Phase 1: Assess, Align & Activate (Days 1–30)

The first 30 days are focused on establishing the operating cadence, aligning internal stakeholders, opening customer discovery tracks, and initiating ecosystem partnerships. No major deals close in this phase — but the quality of work done here determines the velocity of everything that follows.

Internal Alignment: Deep alignment sessions with Google Cloud leadership across sales, product, engineering, sovereign cloud, and partners. Stand up the weekly Global O&G Deal & Product Council. Conduct structured interviews across the organization to understand inherited pipeline quality, product gaps, and competitive dynamics. Produce the first global account heat map scoring all 33 named accounts on technical value, executive sponsorship, competitive friction, and repeat potential. **Customer Engagement:** Open 15 priority executive conversations including Pertamina, PTTEP, Inpex, Reliance, EQT, Devon, Diamondback, and Harbour Energy. Launch formal discovery tracks with Tier 1A targets. Open sovereign engagement tracks with Aramco, KOC, and TotalEnergies. Select 2 fortress accounts and 2 APAC lead accounts for parallel development. **Ecosystem Activation:** Executive workshops with Cognite, SLB, Baker Hughes, and Kongsberg Digital. Open EPAM, Accenture, and TCS/Infosys certification tracks. Initiate Foundation Model co-development scoping with TGS and SLB for first ISV dataset. Qualify initial startup cohort for Google for Startups Cloud Program. Open HUMAIN AI executive engagement for KSA sovereign AI enablement. **Narrative & Think Big:** Publish the agentic OSDU point of view. Lock ADIPEC 2026 keynote slots. Brief DeepMind, Geo, Workspace, Security, and Energy Procurement on module packaging. Brief Google Energy Procurement leadership on Project Interchange. Form SLB × Google Cloud CCUS Steering Committee. **Exit Gate:** Operating cadence ratified; one global account heat map scored; 15 account plans documented; 6+ customer-validated opportunities; 4 ISV JBP drafts; Interchange framework drafted; CCUS Steering Committee formed.

Phase 2: Validate Wedges & Sovereignty (Days 31–60)

The second phase shifts from assessment to active validation — proving the multicloud wedge architecture in live environments, scoping sovereign PoCs with NOCs, and advancing the pipeline to signed proposals.

Fortress Wedge Validation: Launch 2 technical validations (one against an AWS-based estate, one against an Azure-based estate). Demonstrate in-place data access, private connectivity, and Gemini reasoning with agreed production metrics. Document security posture, egress economics, latency benchmarks, and inference cost comparisons. **Sovereign NOC Engagement:** Conduct sovereign design-authority sessions with Aramco/CNTXT, KOC, Pertamina, and PTTEP. Classify each NOC's requirements against the six-region sovereign framework. Deliver Aramco sovereign AI roadmap, KOC hybrid framework, and Pertamina Digital Hub expansion plan. **Pipeline Advancement:** Contract 3–5 Agentic Subsurface Starters with Tier 1A accounts. Scope Devon merger-value workshop and Harbour Energy M&A integration. Open LNG and midstream white space with Williams Kognitwin expansion. Scope TotalEnergies pilot with Cognite and Reliance Jamnagar refinery PoC with AspenTech. **Think Big Advancement:** Draft bilateral MOU template with legal for Project Interchange. Brief DeepMind leadership on Energy Lab concept and identify 1–2 Tier 1 research partners. Deliver ExxonMobil Low Carbon Solutions executive briefing for CCUS. Conduct East Coast Cluster workshop with BP/Equinor/TotalEnergies. Engage INEOS/Greensand. **Exit Gate:** 2 validated multicloud wedge architectures; 1 NOC PoC scope in progress; 3+ accounts signed; Interchange MOU template drafted; DeepMind leadership briefed; CCUS executive briefings delivered to all 3 target consortiums.

Phase 3: Scale, Accelerate & Penetrate (Days 61–90)

The final phase industrializes successful patterns, certifies partners, locks marquee announcements for ADIPEC 2026, and signs the first Think Big MOUs.

Industrialize: Convert the strongest independent launchpad into a repeatable field offering. Certify EPAM and Accenture as anchor GSIs. Publish blueprints, evaluation suites, and governance controls for all 6 agent patterns. Publish Foundation Model validation results. List ≥ 1 startup on Google Cloud Marketplace. **Pipeline Closure:** Secure production-expansion language for successful fortress validations. Formalize 1 sovereign NOC PoC scope with joint architecture sign-off. Sign 1 LNG pure-play. Scale Tier 1 to 10+ qualified pursuits via co-sell. Target: 2 production commitments; pipeline $\geq 4\times$ coverage; $\geq 30\%$ partner-originated. **ADIPEC 2026 Preparation:** Finalize marquee announcements: sovereign NOC story, multicloud fortress proof, ISV co-innovation reveal. Complete customer approvals, legal reviews, and demo scripts. Lock 2–3 customer-approved announceables 30 days pre-event. **Think Big Closure:** Present Project Interchange framework at ADIPEC 2026 as marquee announcement. Announce DeepMind Energy Lab with 1 signed Tier 1 research partnership. Secure ≥ 1 CCUS consortium MOU/LOI. Deploy MVP MRV Agent sandbox. Publish Broadwing CCS reference architecture. Position all three initiatives for CERAWEEK 2027. **Exit Gate:** 3 solution kits shipped; GSI certification complete; 1 FM validated; 1 Interchange MOU signed; 1 DeepMind partnership signed; ≥ 1 CCUS MOU signed; all three Think Big initiatives announced at ADIPEC; 90-day board review delivered; FY27 plan endorsed.

Part V: Strategic OKRs

Six objectives define our success criteria for this 90-day plan:

Objective 1 — Establish Google as the agentic-AI leader in oil & gas. Key Results: 3–5 Tier 1 accounts signed with ≥ 1 live in production on Agentic OSDU; 1 PE sponsor-platform framework in negotiation; 6-agent reference portfolio published with 2–3 customer-approved announceables for ADIPEC; 1 GCP-hosted industry foundation model in partner validation with ≥ 1 ISV dataset. **Objective 2 — Prove the multicloud wedge in fortress accounts.** Key Results: 2 wedge validations live (1 AWS, 1 Azure) with signed success criteria; ≥ 1 advanced to production-commitment negotiation; zero migration-framing escalations. **Objective 3 — Unlock the sovereign Gulf & APAC opportunity.** Key Results: 1 Gulf NOC PoC scope ratified and joint architecture signed; sovereign framework validated across 6 regions; Pertamina/PTTEP expansion live; HUMAIN AI PoC scope ratified with Aramco. **Objective 4 — Build an ecosystem that outsells the direct force.** Key Results: 3 ISV charters signed; 2 anchor GSIs certified; $\geq 30\%$ of qualified pipeline partner-originated; ≥ 3 startups on GCP with ≥ 1 on Marketplace. **Objective 5 — Convert the Alphabet advantage into commercial gravity.** Key Results: 3 Alphabet modules embedded in live pursuits; TotalEnergies reciprocity pursuit active; 2 CEO-level Energy Exchange dialogues opened. **Objective 6 — Launch THINK BIG: Interchange, DeepMind Lab & CCUS.** Key Results: 1 Interchange MOU signed; 1 DeepMind research partnership signed; ≥ 1 CCUS consortium MOU signed; all three announced at ADIPEC 2026.

Part VI: Executive Decisions Required

This strategy requires several explicit leadership decisions to proceed. These are not operational details — they are structural choices that define the boundaries of what the team can execute:

Confirm the 33-account portfolio and resource-concentration model. Ratify P1 direct ownership versus P2/P3 partner-led with review gates. Assign named executive owners for each P1 account.

Name product and engineering owners for the six reference agents and three Alphabet modules. Each agent pattern (Subsurface, Well Log, Drilling Risk, Production, Reliability, Process Safety) and each Alphabet module (Geospatial, WeatherNext, Security) requires a named Google owner.

Approve the multicloud co-existence architecture as an explicit, sanctioned sales motion. Formally authorize BigQuery Omni + Cross-Cloud Interconnect + Gemini-over-existing-estate as the standard fortress/wedge account play. This is required before sellers can position it with Shell, Oxy, Chevron, Equinor, and others.

Ratify the sovereign framework as a unified NOC decision model across 6 regions — not six separate products. This requires alignment across sovereign cloud, sales, and legal teams.

Authorize the Foundation Model co-development framework with ISV data partners. Confirm the zero-joint-IP policy between Google and ISVs and approve Vertex AI hosting architecture with Trillium TPU / A3 Ultra GPU compute allocation for FM pre-training.

Authorize Project Interchange with Google Energy Procurement. Approve the bilateral framework, identify pilot accounts, and establish legal decoupling of power off-take and cloud licensing.

Authorize DeepMind Energy Lab. Approve the structured research program, scope Tier 1 partnerships (\$10–50M multi-year), and constitute the academic advisory board (Rice Baker Institute, KAUST).

Authorize CCUS Transformation Partnerships. Approve the SLB × Google CCUS Steering Committee, authorize engagement with ExxonMobil LCS, East Coast Cluster, and Greensand, and approve MVP MRV Agent sandbox deployment.

Authorize ADIPEC 2026 and CERAWEEK 2027 announcement workstreams. Approve 2–3 customer-approved announceable targets and demo builds for both events.

Approve the Google for Startups Cloud Program cohort and HUMAIN AI sovereign enablement. Confirm the startup cohort for up to \$200k credits and authorize HUMAIN AI platform enablement on Damman me-central2.

GTM Operating Model

The execution of this strategy depends on a cross-functional operating model that connects Sales, Business Development, Solutions Engineering, GTM & Marketing, and Engineering & Product into a coordinated system. The operating rhythm includes:

- **Weekly Global O&G Deal Council** — Pipeline review, deal strategy, blocker removal, resource allocation, competitive intelligence sharing.
- **Bi-weekly Product & Engineering Sync** — Field signal translation into product roadmap; feature requests, PoC learnings, sovereign requirements, gap analysis.
- **Bi-weekly Partner Ecosystem Review** — JBP progress, co-sell pipeline, certification status, partner escalations.
- **Monthly GTM Campaign War Room** — Event prep, demand-gen performance, analyst/media relations, content calendar.
- **Monthly Executive Briefing Council** — Account health, competitive intelligence, strategic pivots, resource asks.

- **Quarterly SA & Solutions Bootcamp** — Deep-dive training on new agent patterns, sovereign architectures, competitive updates, hands-on labs.

Field enablement is tracked through nine purpose-built kits: Account Attack Playbook, Agentic OSDU Battle Card, Multicloud Wedge Pitch Kit, Sovereign NOC Decision Framework, Partner Co-Sell Playbook, ADIPEC Event Playbook, HPC & Seismic Win Kit, Industry Foundation Models Sales Kit, and Startup & HUMAIN AI Ecosystem Kit.

Data Sources & References

All market figures, competitive positioning, and industry statistics in this document are sourced from the following publicly available research and institutional publications:

- [IEA World Energy Outlook 2025](#) — Global energy demand, revenue benchmarks, and electricity consumption forecasts.
- [Gartner IT Spending Forecast Q2 2025](#) — Vertical IT expenditure metrics and cloud adoption rates by industry.
- [IDC Cloud Tracker 2025](#) — Enterprise public cloud adoption rates and market share analysis.
- [Precedence Research 2025](#) — AI in Oil & Gas market projection (\$5.4B today, \$18.7B by 2035).
- [MarketsandMarkets 2025](#) — Digital Oilfield market forecast (\$37B today, \$43B by 2029).
- [Global CCS Institute Status Report 2025](#) — Global CCUS project tracker and investment analysis.
- [SPE Journal of Petroleum Technology](#) — Upstream geoscience workforce surveys and digital energy conference proceedings.
- [Saudi NCA Regulations](#) — National data residency and sovereign cloud compliance requirements.
- [Everest Group Energy Cloud Analysis](#) — Cloud market share analysis for the energy vertical.

Account-level opportunity estimates, competitive spend figures, and revenue projections represent directional analysis based on public disclosures, analyst reports, and internal pipeline assessment. Actual GCP spend data is managed via CRM/Salesforce records.

This document is intended for internal strategic planning purposes only. Market data includes directional estimates informed by publicly available sources. All competitive positioning represents Google Cloud's analytical assessment and does not reflect proprietary competitor information.